

MUCHUKUNTLA BRAHMANAIDU

Bangalore, India — brahmanaidu2172@gmail.com — +91 8688064974

LinkedIn — GitHub — Portfolio

Professional Summary

Machine Learning Engineer specializing in **MLOps**, **Computer Vision**, and **NLP**. Experienced in architecting **end-to-end ML pipelines**, deploying **scalable microservices** with **Docker** and **FastAPI**, and optimizing **model inference** for production. Proven track record of implementing **FAISS** vector search, reducing **inference latency** by 40%, and building **CI/CD** workflows for automated model deployment. Skilled in **PyTorch**, **TensorFlow**, and **cloud-native ML infrastructure**.

Technical Skills

- **MLOps & Deployment:** Docker, FastAPI, REST APIs, Model Serving, CI/CD, ONNX Runtime, Microservices Architecture, API Gateway
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Scikit-learn, CNN, ResNet, Transformers, Neural Networks, Large Language Models (LLMs), Computer Vision, NLP, Model Optimization
- **Vector Search & Retrieval:** FAISS, Vector Databases, Similarity Search, Embedding Pipelines, Semantic Retrieval
- **Data Engineering:** SQL, PostgreSQL, Pandas, NumPy, ETL Pipelines, Data Preprocessing
- **Cloud, DevOps & Languages:** AWS, Vercel, Git, GitHub, Linux, Python, R

Professional Experience

Zithara.AI (Propel5000)

Hyderabad, India

Machine Learning Engineer Intern

January 2025 – April 2025

- Architected **end-to-end MLOps pipeline** for a **visual search system**, enabling similarity-based retrieval across a 5,000+ SKU jewelry catalog using **PyTorch ResNet50** embeddings and **FAISS** vector indexing
- Deployed **containerized ML microservice** using **Docker** and **FastAPI**, serving 100+ concurrent API requests with **sub-150ms inference latency** and 99.9% uptime
- Implemented **ONNX model conversion** and **batch inference optimization**, reducing model loading time by 40% and memory footprint by 35%
- Built automated **CI/CD pipeline** for model versioning and A/B testing, enabling seamless production updates without service interruption
- Designed **embedding pipeline** for live feature extraction, achieving 94% precision in design texture recognition and visual similarity matching

Projects

Vision Guard – Real-Time Defect Detection System

Python, PyTorch, OpenCV, ONNX, Docker

- Developed **production-grade computer vision pipeline** using **CNN/ResNet** architectures for industrial defect detection, attaining **96% classification accuracy** on 10,000+ test images
- Optimized **model inference** to **sub-140ms latency** using **ONNX Runtime** and **batch processing**, enabling real-time industrial inspection at 30 FPS
- Deployed **horizontally-scalable microservice** with **Docker** containerization, improving defect detection accuracy by 92% in production trials

Smart Hire AI – Neural Resume Screening Platform

NLP, Sentence Transformers, FastAPI, Docker, CI/CD

- Built **AI-powered screening system** achieving **90% semantic alignment** between resumes and job descriptions using **Sentence Transformers** and **Cosine Similarity**
- Automated HR workflow, reducing manual review time by **80%** (from 5 hours to 1 hour per 100 candidates) through a **multi-stage LLM pipeline** and intelligent ranking
- Engineered **skill gap analysis module** using **transformer-based embeddings** to rank 500+ candidates by semantic relevance and qualification match

Market Pulse – Customer Segmentation & RFM Analysis

Python, Pandas, Scikit-learn, Power BI

- Developed a **full-stack ML pipeline** using **KMeans clustering** and **RFM Analysis** (Recency, Frequency, Monetary) to identify 5 distinct customer segments from 10,000+ records
- Achieved **Silhouette Score of 0.72**, demonstrating strong cluster cohesion and delivering actionable insights for targeted marketing campaigns
- Generated **data-driven retention strategies** identifying a potential **15% increase in customer retention** through personalized segment outreach

Education

Bachelor of Technology (B.Tech) in Computer Science

CGPA: 7.9/10

Annamacharya Institute of Technology and Sciences (JNTUA)

December 2021 – April 2025

Certifications

- Google Data Analytics Professional Certificate – Coursera (SQL, Tableau, R, Data Pipeline)
- Data Science and Machine Learning Bootcamp – Udemy (Python, Scikit-learn, Neural Networks)